Tutorial 2

Cauchy's Integral Theorems

Using Cauchy's integral theorems, evaluate the following contour integrals.

1. $\oint_C \frac{z^2 - 4z + 4}{z + i} dz$ where C is the circle $|z| = 2$.
2. $\oint_C \frac{z}{z^2 + 9} dz$ where C is the circle $|z - 2i| = 4$.
3. $\oint_C \frac{z + 1}{z^4 + 2iz^3} dz$ where C is the circle $|z| = 1$.
4. $\oint_C \frac{1}{z^2 + 2z + 2} dz$ where C is the circle $|z - i| = 2$.
5. $\oint_C \frac{1}{z^2 + 2z + 2} dz$ where C is the circle $|z| = 2$.
6. $\oint_C \frac{\ln z}{z^2 + iz - 1} dz$ where C is the circle $|z - 1| = \frac{3}{5}$.
7. $\oint_C \frac{e^{\alpha z}}{z^2 + iz - 1} dz$ where C is the circle $|z| = 2$.
8. $\oint_C \frac{\ln z}{z^2 - 2z + 2} dz$ where C is the circle $|z - 2| = \sqrt{3}$.
9. $\oint_C \frac{e^{\alpha z}}{(z^2 + 1)^2} dz$ where C is the circle $|z| = 2$.
10. $\oint_C \frac{\ln(z + 1)}{z^{n+1}} dz$ where C is the circle $|z| = \frac{1}{2}$.

Additional problems:

Zill & Shanahan:
Ex 5.3, p.262-263, Q 1-20 (odd numbers)
Ex 5.5, p. 281-282, Q 1-22 (odd numbers).